

Linear systems and elimination

Sharp growth for cyclic tridiagonal matrices

Difficulty: challenging **Importance:** interesting to specialist

Status: [Lean verified](#)

Lean verification — 22 September 2026

The complete original target is formally proved: for every $n \geq 4$, the sharp growth factor is $c_n = F_{n+1} + 1$. The upper bound covers every nonsingular complex cyclic tridiagonal input with both cyclic corners nonzero, every permitted GEPP tie path, and every active Schur entry in the original ordering. The explicit rational family attains the bound through actual row swaps. The nonempty bounded growth set, its greatest member and its genuine real supremum are all proved.

All four exports passed LeanCert kernel audits, two independent nonauthor final reviews, and [isolated Linux Comparator/default-kernel verification](#) with actual rejection and sandbox controls. [Proof and reviews](#) · [Immutable proof revision](#) · [Retained evidence](#).

Mathematics: Matthew J. Colbrook; Higham retains original-problem credit. Formalization: Sidney Holden with OpenAI Codex assistance. No symmetry, diagonal dominance, real-only restriction or deterministic tie rule is substituted. No novelty, source-author endorsement or external human peer review is claimed. The complete original target and earlier source record remain below.

Independently reviewed resolution - 2026-09-11

Sharp growth classification. Theorem 1 and Sections 2-4 prove $c_n = F_{n+1} + 1$ for every $n \geq 4$, with $F_0 = 0, F_1 = 1$. The bound covers complex cyclic tridiagonal matrices, every active entry and every permitted GEPP tie path in the original ordering. A rational matrix with both cyclic corners nonzero attains it at every order.

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The difficulty, importance and rating rationale below are historical assessments of the original open target. Original statements, references and dated audits are preserved.

Rating rationale: Challenging reflects an all-orders extremal problem in which the two corner entries change elimination fill; specialist impact is a precise stability bound for cyclic tridiagonal systems.

Problem statement

For each $n \geq 4$, let C_n consist of the nonsingular matrices $A \in \mathbb{C}^{n \times n}$ whose only permitted nonzero positions satisfy

$$|i - j| \leq 1 \quad \text{or} \quad (i, j) \in \{(1, n), (n, 1)\},$$

with $a_{1n} \neq 0$ and $a_{n1} \neq 0$. This is the cyclic tridiagonal pattern called *quasi-tridiagonal* in the source. No symmetry or diagonal dominance is assumed.

Apply Gaussian elimination with partial pivoting (GEPP) in exact arithmetic. At each stage select a largest-modulus entry in the active first column, interchange its row with the active first row, and form the trailing Schur complement. For an admissible path π , let $S_1 = A, \dots, S_n$ denote the active matrices. Define

$$\rho(A, \pi) = \frac{\max_{1 \leq k \leq n} \|S_k\|_{\max}}{\|A\|_{\max}}, \quad \|M\|_{\max} = \max_{i,j} |m_{ij}|,$$

$$c_n = \sup_{A \in C_n} \sup_{\pi \text{ permitted by GEPP}} \rho(A, \pi).$$

Determine c_n for every $n \geq 4$, with matching upper bounds and examples approaching each supremum. All admissible tie choices are included. The given ordering is fixed before the stated GEPP run; preliminary row or column reorderings are not permitted. All sizes constitute one extremal problem.

Ordinary tridiagonal inputs have growth at most two. This question isolates the effect of the two corner entries on partial-pivoting stability. Unlike the fixed-bandwidth problem IE-13, the corner offsets grow with n .

Reference

N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, second edition, SIAM (2002), Problem 9.15(b), p. 193; Theorem 9.11, p. 173, supplies the ordinary tridiagonal comparison.

Earlier status check — 2026-09-08

The book's sparsity pattern was checked directly. Its wording omits the field; this entry adopts $\mathbb{C}$ from the adjacent Theorem 9.11. Searches for *quasi-tridiagonal growth factor*, *quasi-tridiagonal partial pivoting*, and *Gaussian elimination cyclic tridiagonal growth* found no solution of this exact extremal question. A [2017 cyclic-reduction paper](#) uses a different matrix pattern and algorithm. Shah–Urschel's [August 31, 2026 revision](#), Theorems 2.2–2.3, does not determine this extremum. No recent explicit reaffirmation of the question's openness was found; absence of a located resolution is the limit of this check.

Audit update — 2026-09-10

Rechecked Higham's [Problem 9.15\(b\)](#), p. 193, against the two-corner pattern displayed here. Searches for cyclic/quasi-tridiagonal pivot growth and later elimination bounds found no exact solution. The [August 2026 general complete-pivoting results](#) address a different pivot rule and do not provide this pattern's sharp partial-pivoting bound; current openness rests on the historical question and this limited search.